BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Hardware

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is the role of a server in a client-server network?

2. When does iterative testing happen?
   A) Only after delivery
   B) Only before any coding
   C) During development
   D) Only during installation

3. For valid integer ages 13-18, give an invalid input of the correct type.

4. A public booking form takes text and numbers. "Defensive design assumes every input is valid and safe." Correct this. What could happen?

Today's objectives

Describe the roles of a NIC, switch, router and WAP. Distinguish local MAC-based forwarding from routing between networks.

Key vocabulary - Read, cover, recall

NIC
Network Interface Card; connects a device to a network.

Switch
Connects LAN devices and forwards traffic using MAC addresses.

Router
Forwards data between networks using IP addresses.

WAP
Wireless Access Point; provides wireless connection to a network.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

NIC

A Network Interface Card (NIC) lets a computer connect to a network and transmit/receive data. Interfaces have MAC addresses.

Switch

A switch connects devices in one LAN and forwards local data using MAC addresses.

Router

A router moves data between different networks, using destination IP addresses.

Wireless access point

A WAP connects wireless devices to a network, normally through the existing wired LAN.

I Do - Worked model

A student sends coursework to a printer on the same school LAN. Explain what the NIC and switch do. [4 marks]

- The computer and printer are in the same LAN; the data must leave the computer.
- A NIC provides network connectivity [1] and sends the data [1].
- The switch connects LAN devices [1] and forwards the traffic using MAC addresses [1].
- Two distinct devices with two accurate marking points each.

We Do - Guided practice

1. A pupil visits a website outside the school network. Explain the roles of the NIC and router. [4 marks]

2. Tablets must access a school's existing wired network. Explain why a WAP and switch are needed. [4 marks]

3. Explain two differences between a router and a switch. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: What prints when local is false?

```
local = false
if local then
print("Stay on LAN")
else
print("Use router")
endif
```

You Do - Independent exam questions

Q2. State what a NIC allows a computer to do. [1 mark]

Q3. A supermarket introduces wireless barcode scanners that must access its existing wired LAN. Explain the role of a WAP and a switch. [4 marks]

Q4. A museum information kiosk requests a catalogue page hosted on another network. Explain the roles of its NIC, the local switch and the router. [6 marks]

Exit Ticket - True or False?

A router forwards between networks using IP addresses.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It provides centrally managed services or resources requested by clients.

2. C - During development

3. 12 is an integer but outside the valid range, so it must be rejected.

4. Plan for invalid, unexpected or harmful input and handle it safely. A known user can still enter unsuitable data.

I Do - Explain the model

The computer and printer are in the same LAN; the data must leave the computer.

A NIC provides network connectivity [1] and sends the data [1].

The switch connects LAN devices [1] and forwards the traffic using MAC addresses [1].

Two distinct devices with two accurate marking points each.

We Do - Model answers

1. The NIC connects the pupil's computer to the LAN [1] and sends/receives traffic [1]. The router forwards data between the LAN and external networks [1] using IP addresses [1].

2. WAP lets tablets join wirelessly [1] by radio [1]. Switch links the WAP to the LAN [1] and forwards data to intended LAN devices [1].

3. A switch handles traffic within a LAN [1] using MAC addresses [1]. A router forwards between networks [1] using IP addresses [1].

Code First - Answer

```
local = false
if local then
print("Stay on LAN")
else
print("Use router")
endif
```

Expected output: Use router

You Do - Self-mark

Q2: Connect to a network to send/receive data [1].

Q3: WAP connects wireless devices [1] using radio [1]. Switch links wired devices/WAP in a LAN [1] and forwards using MAC addresses [1].

Q4: NIC connects device [1] and transmits data [1]; switch connects LAN devices [1] using MAC forwarding [1]; router connects networks [1] using IP addressing [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.